



KJ 292

UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject: PHYSICS

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

Item 1 EXPECTED RESPONSE

a(i) Mirror A is a concave mirror, The image appears bigger when the person stands between the focal point and surface/pole of the mirror.

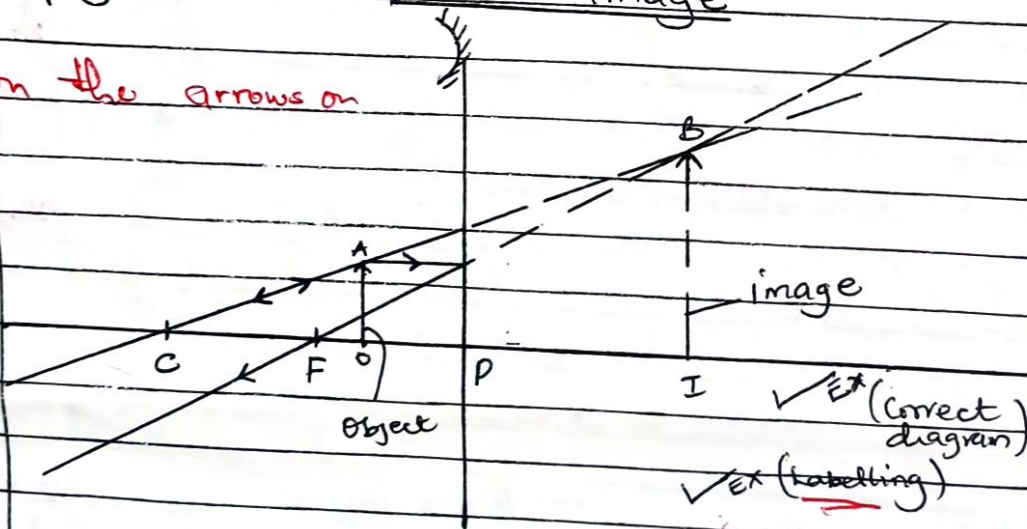
This makes the reflected rays diverge. When the reflected / diverging rays are extended backwards they appear to converge at a point behind the mirror creating an image that is magnified upright and virtual image.

REMARKS

- Concave mirror ✓
- Point P ✓
- extend rays ✓
- converge ✓
- upright ✓
- virtual ✓
- lateral inversion ✓
- Any two properties score ✓
- Ex ✓

Any two properties of image formed.

- Relax on the arrows on rays



a(ii)



$$V = \frac{2d}{t}$$

✓ (information gathering)

Information gathering means sketching and identifying the required distance
kd



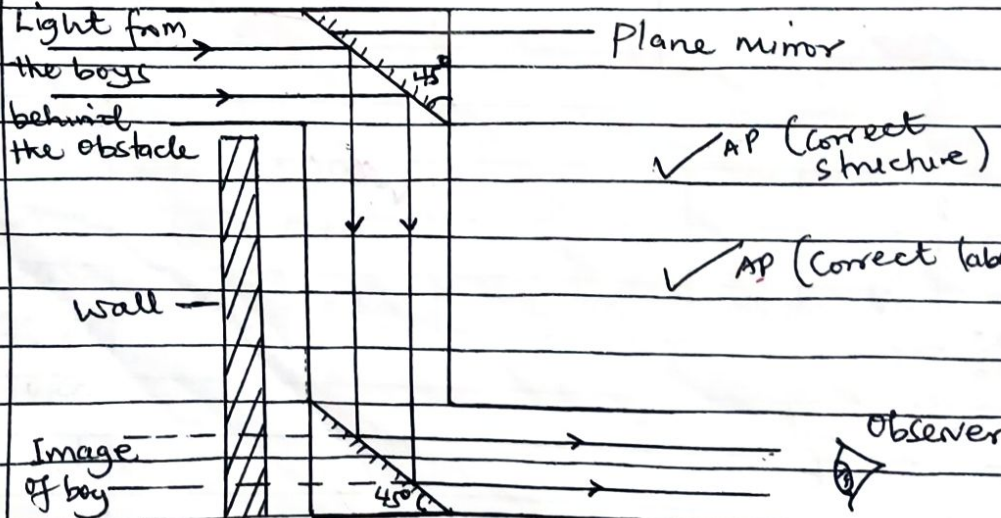
UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Year: 2025

Level: ☐ PLE ☒ UCE ☐ UACE

	EXPECTED RESPONSE	REMARKS
$V = \frac{2d}{t}$	AN (Substitution)	
$330 = \frac{2d}{0.5}$	$330 = 2(60 + 8 + Kd)$ ✓	Substitution ✓ AN
$d = 82.5$	0.5	Value obtained ✓ AN
$d = 82.5$	$82.5 = 68 + Kd$	Conclusion ✓ AN
$d = 82.5 - (68)$	AN (value obtained)	
$= 14.5m$	$Kd = 14.5m$ ✓	
	AN (Conclusion)	
	The wall will not be broken down as it is 14.5m from the road far greater than the minimum 10m required.	
	b) The mirror B is a plane mirror. The optical instrument to be designed by Kyle is a simple periscope. ✓ AP	Plane mirror ✓ AP Simple periscope
	Two plane mirrors are fixed at corners of a pipe parallel and facing each other to receive rays at an angle of incidence of 45° ✓ AP	Arranged mirrors ✓ AP
		Correct diagram ✓ AP A plane mirror ✓ AP Parts AP ✓ AP



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

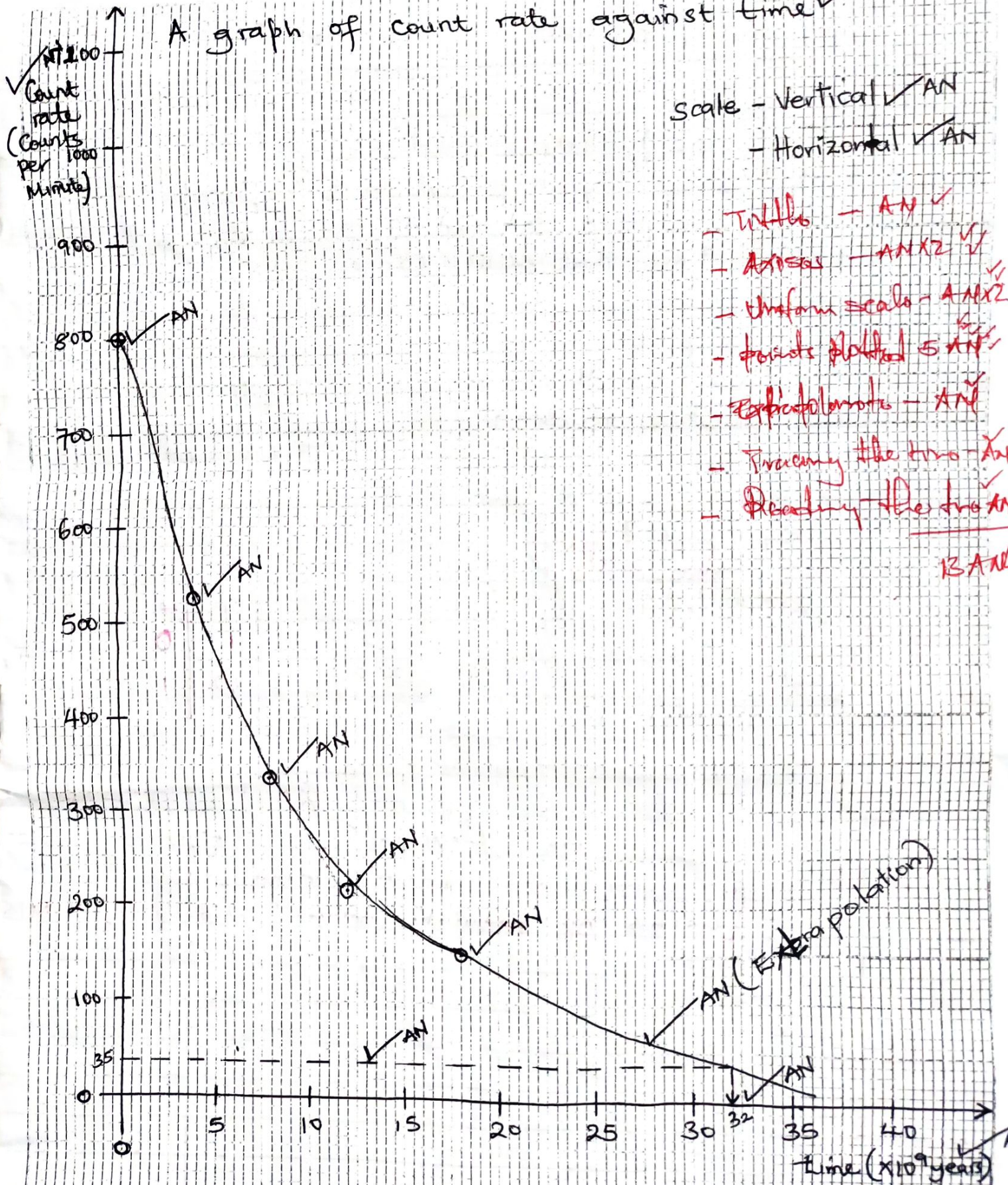
Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

	EXPECTED RESPONSE	REMARKS
Health? danger? Radiation? Any 3 dangers	Item 2 a) There is need to relocate in order to avoid effects/dangers when exposed to radiations like alpha particles, beta particles and gamma rays emitted by the radioactive material as it decays. The dangers anticipated include; - Skin irritation ✓ EX - Skin sores ✓ EX - Cancer ✓ EX - Reduced body immunity - Eye cataracts - Genetic mutation - Sterility From the graph time taken $t = 32 \times 10^9$ years have to elapse before they return $[(32 \pm 5) \times 10^9 \text{ years}]$	The radiation levels are so high above the background radiation which when exposed lead to effect OR; From Graph $T_{1/2} = 800 \rightarrow 400 \rightarrow 200$ $25 \leftarrow 50 \leftarrow 100$ time = $5T_{1/2}$ $= 6.5 \times 5$ $= 32.5 \times 10^9 \text{ y}$ Conclusion - AN
Identify + purpose or identify + process	2(b) Generation of electricity in nuclear power plant. Heat produced from nuclear reaction fission heats the water to produce steam at high pressure. The steam runs the turbines connected to generator producing the electricity	chromium oxide - Heat ✓ AP - Fusion ✓ AP

A graph of count rate against time ✓ AN



BANK



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

Used in hospitals / medical field to treat cancer. The affected part of the body is exposed to gamma rays for a short time that kills cancer cells.

Accept any two explained applications.

- detection of leakages in oil or water pipes
- Used as thickness gauge in paper or plastic industries
- Detecting bone fractures
- used to sterilise medical equipment.

Carbon dating

Code	Weighted score		6	5	4	4	4
PC ₂	EX		3 and above	2	1	1	0
	AN		9 and above	5-8	1-4	0	1
PA ₂	Weighted score		6	4	3		
	AP		3 and above	2	1		

Score *Application* + *purpose*
or score in total how the process of *Application* occurs (states place) marks



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

Item 3 EXPECTED RESPONSES

3a i) The water behaviour observed is as a result of ocean tides. These are the rise and fall of sea water levels. They are caused by gravitational pull of the moon and the sun combined with the earth's rotation. The moon's gravity creates bulges of water on the side of earth facing the moon and the opposite side resulting in high tides. As the earth rotates, different locations move through the bulges causing the cyclical nature of tides.

ocean tides - Ex.
Rise and fall of sea water - Ex.
Gravitational pull - Ex.
Bulges of water (high tides) - Ex.
Side facing the moon - Ex.
Opposite side - Ex.

3a ii) The Google link application sent by her friend works by the help of Global Positioning System (GPS) technology. It uses communication / Navigation satellites to locate positions.

GPS - Ex.
Navigation / Comm - Ex.
Transmits signals - Ex.
How long - Ex.
Trilateration / 3-dimensional - Ex.
Distance

A GPS receiver picks up signals from satellites. These receivers determine how long the signal takes to be received from the satellites and compute the distances from the different satellites. By computing the distance from each satellites receiver determines the accurate location on earth in 3 dimensions in terms of latitude, longitudes and altitudes.



UGANDA NATIONAL EXAMINATIONS BOARD DRAFT

Subject:

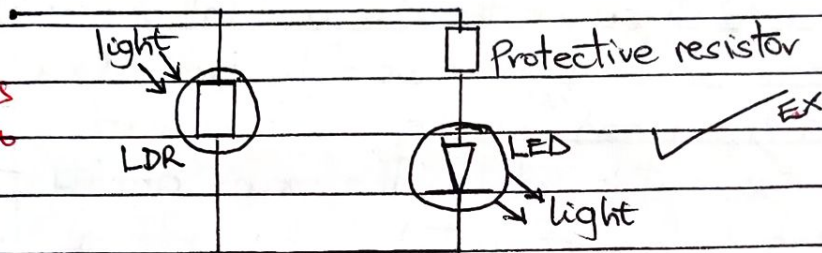
Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

3 b) The circuit works by removing the NOT gate ✓EX and completing the circuit.

Removing NOT gate



Light dependent resistor ✓EX

Resistance

increases with intensity ✓EX

The security light uses the light dependent resistor (LDR). The resistance of the LDR increases as the light intensity reduces. At night the resistance increases and the current flow in the LDR is limited. Most of the current passes through the LED causing it to light.

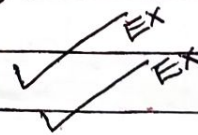
Resistance increases ✓EX

Current flow in LDR limited

The truth table of a NOT gate.

Total Ex's - 08

A INPUT	OUTPUT B
0	1
1	0



3 c) Navigation and Shipping: Accurate tidal predictions are necessary for safe marine navigation. The tides determine water depths in ports and shallow channels, allowing large cargo ships to safely enter and leave at appropriate times.

Accept any two applies of ocean tides explained accordingly

-Renewal energy generation (Tidal energy generation)

-Tidal carry water rich nutrients.



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

- ✓
- Used for recreation.
 - Coastal engineering. Understanding tidal forces and currents are essential for the design and construction of coastal infrastructure such as bridges, sea walls to ensure their stability and resilience.

Weighted score		6	5	4	3
SC ₃	EX	11 and above	6-10	3-5	1-2
	AN				
Weighted score			6	5	4
SA ₃	AP		3 and above	2	1

~~3~~ 3

✓



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

EXPECTED RESPONSES

Item 4

4a) The friction force between the moving parts of the bicycle causes persistent and discomforting noise. It can be minimised by lubricating/oiling the parts in contact and move relative to each other.

Frictional force ✓
lubricating ✓
replacing ball bearing
tightening the loose
nuts(parts) / Aligning
the gear system.

b) Total distance = 10km = 10,000m

After covering 6km, 4km remains = 4000m

At 6km covered and 7:40am initial

Speed = 5ms^{-1} final speed = 15ms^{-1}

Using $v^2 = u^2 + 2as$

$$15^2 = 5^2 + 2 \times a \times 4000$$

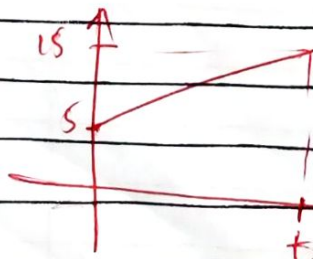
$$225 = 25 + 2 \times a \times 4000$$

$$225 - 25 = 8000a$$

$$a = \frac{200}{8000}$$

$$= 0.025\text{ms}^{-2}$$

The rate of change of velocity is the acceleration



$$\frac{1}{2} h(a+b) = 4000$$

$$\frac{1}{2} t(5+15) = 4000$$

$$v = u + at$$

=



UGANDA NATIONAL EXAMINATIONS BOARD

DRAFT

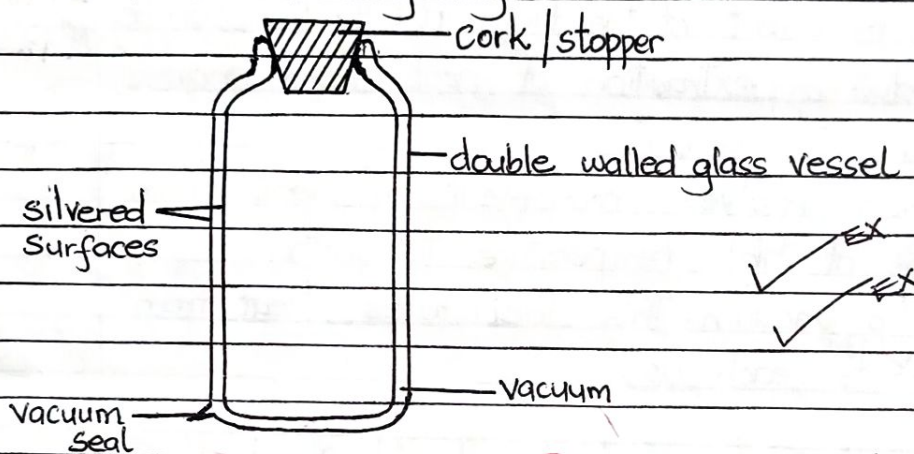
Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

4c) He should use a vacuum flask that helps to keep the tea warm in many ways:-



Heat transfer by conduction and convection is minimised by making the flask a double walled glass vessel with a vacuum between the walls. Radiation is reduced by help of silvered walls on both sides of the vacuum.

Conduction is further reduced by using a poor conducting material as the stopper which also minimises heat loss by evaporation.

4d) Application of conduction, convection and Radiation.

- Conduction. Heat conduction process is used in good conductors like sauce pan materials or food warmers/boilers to enable heat transfer from the source of heat to where it is required.

Conductors both good and poor conductors applications

Conduction / convection
Heat loss
Two is more

Accepts any one application of each of the processes explained accordingly



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

- Convection: This is a method of heat transfer through fluids. Heat moves through a fluid from area of high temperature to area of low temperature by actual movement of the fluid itself. This can be applied during extraction of heat from engines by radiators by convection.

- Radiation involves movement of heat from one place of high temperature to another through a vacuum. This facilitates heat from the sun to reach us.

*Convection - Plan
formal
Vandolentel
- coolay food
- Radiator -
- Food preservation
- Green house
- Solar energy*

Weighted score		6	5	4	2	2	3
TC ₄	EX	9 and above	6-8	3-5	1-2	2	3
	AN	4 and above	3	2	1	1	0
TA ₄	weight score		6	5	4		
	AP		3	2	1		

*Candidates with EX - 00 } weight 2
AM - 03 }*

*Slow weight
EX 1-2 }
AM 01 }*

** EX - 1 } weight 2
AM - 0 }*

Note In case of a minimum



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Paper code: 535/1

Subject:

Year: 2025

Level: ☐ PLE ☒ UCE ☐ UACE

EXPECTED RESPONSES

Item 5

5a) The wall was designed with thicker walls at the bottom and wider at the base in order to have a stronger foundation / stable base that can withstand the greater pressure caused by a large weight of the water / Pressure is higher at deeper ends or heights.

- Stronger foundation / stable ✓ Ex
- withstand pressure ✓ Ex
- water pressure is higher ✓ Ex

5b) The steel bars meet requirements for use if the ratio of stress to strain exceeds $2.1 \times 10^{11} \text{ Nm}^{-2}$

$$r = \frac{2.5 \times 10^{-2} \text{ m}}{2} \quad \text{AN (radius)} \quad \text{AN (conversion)}$$

$$A = \pi r^2$$

$$= 3.14 \times \left(\frac{2.5}{2} \times 10^{-2} \right)^2 \quad \text{AN}$$

$$= 4.90625 \times 10^{-4} \text{ m}^2 \quad \text{AN}$$

Radius ✓ AN
Conversion ✓ AN
Area ✓ AN
Area ✓ AN
Substitution ✓ AN
Stress ✓ AN

Stress = force

area of cross section

$$= \frac{7.7 \times 10^5}{4.90625 \times 10^{-4}} \quad \text{AN}$$

$$= 1.57 \times 10^9 \text{ Nm}^{-2} \quad \text{AN (value)}$$

- Area of stress ✓ AN
Substitution ✓ AN
Stress ✓ AN

- Answer of stress ✓ AN

- Ratio of stress ✓ AN

- Answer ✓ AN

$$\text{Strain} = \frac{\text{extension}}{\text{original length}} = \frac{0.2 \times 10^{-3}}{3.2} \quad \text{AN}$$

$$= 6.25 \times 10^{-5} \quad \text{AN}$$

- Conclusion ✓ AN



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Paper code: 535/1

Subject:

Year: 2025

Level: ☐ PLE ☒ UCE ☐ UACE

$$\begin{aligned} \text{Ratio of stress to strain} &= \frac{1.57 \times 10^9}{6.25 \times 10^{-5}} \text{ AN} \\ &= 2.511 \times 10^{13} \text{ Nm}^{-2} \text{ AN} \end{aligned}$$

The steel bars were of good quality since the ratio of stress to strain is $2.511 \times 10^{13} \text{ Nm}^{-2}$ that exceeds the minimum required of $2.1 \times 10^{11} \text{ Nm}^{-2}$.

Sea breeze ✓
- cooling side ✓

5c) The breeze they experienced was a sea breeze.

Applications of the sea breeze

- Cooling coastal cities, ventilating building.

The breeze helps in cooling coastal cities and improving air quality by blowing in cool air and disperse air pollutants to improve air quality.

- Renewable energy and weather forecasting powering wind mills. Coastal wind mills are specifically designed to take advantage of the regular wind speeds from sea breezes to generate electricity.

Accept any two applications of the sea breeze explained accordingly

- Renewable ✓

Weighted score		6	5	4	2
TC ₅	EX	3	2	1	1
	AN	8 and above	6-7	3-5	1-2
weighted score			6	5	4
TA ₅	AP	3 and above	2	1	

2/2
1/0
0/1



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

15

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

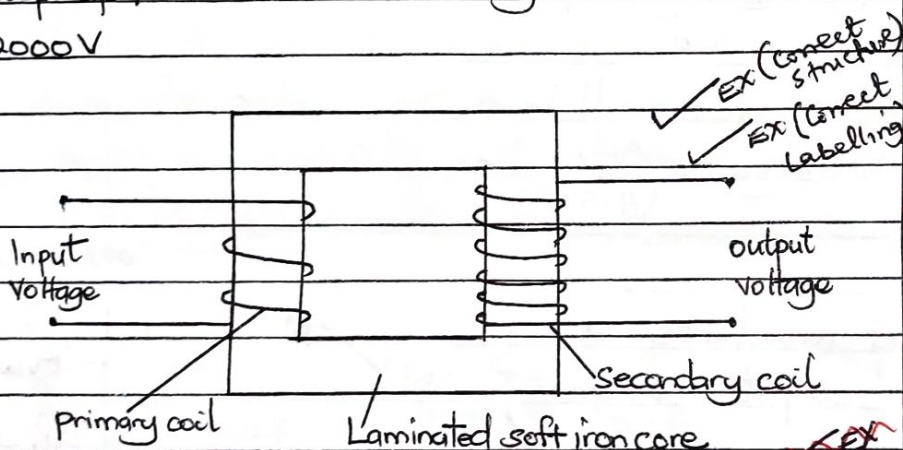
Year: 2025

EXPECTED RESPONSES

Item 6

6 a) The device needed to sort out the voltage issue is a step up transformer, so that it will step up / increase the voltage from 415V to 2000V

- Step up ✓ Ex
- Transformer ✓



- Diagram ✓
Ns > Np
- two labelling ✓

A step-up transformer has a primary coil and a secondary coil wound onto a laminated soft iron core. It has more number of turns on secondary coil than on the primary coil.

- It has more Ns > Np ✓
- Sort iron ✓

The coils are wound either on top of one another or on separate limbs of the core.

- mutual induction ✓
- Alternating / changing current ✓

The transformer uses the principle of mutual induction.

- EMF induced ✓

When an alternating voltage is connected to the primary coil an alternating current flows through the primary coil causing a changing magnetic flux which links the secondary coil.

- showing Ns > Np ✓

An emf is induced in the secondary coil whose magnitude is determined from the transformer equation.



UGANDA NATIONAL EXAMINATIONS BOARD

DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\text{from } \frac{2000}{415} = \frac{N_s}{N_p} \quad \checkmark \text{EX}$$

$$N_s = \left(\frac{2000}{415} \right) N_p$$

The design required will have the number of turns on secondary = $\left(\frac{2000}{415} \right)$ times that on primary.

6b) Cable resistance 2Ω

Power output; 2kW = 2000W $\checkmark$ AN (conversion)

$$P = VI$$

$$2000 = 2000I$$

$$I = 1A$$

Power loss in cables $P = I^2 R$

$$= 1^2 \times 2$$

$$= 2W$$

The device that transforms 415V to 2000V for transmission is efficient $\checkmark$ because it limits power loss to 2W which is less than maximum 5W.

- Conversion kW to W $\checkmark$

Substitution $\checkmark$

$$P = VI \quad \checkmark$$

Power Loss $\checkmark$

Substitution $\checkmark$

Ans 5 turns $\checkmark$

Conclusion $\checkmark$



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

✓ 6c) - Ohmic losses: - This is energy lost in form of heat due to the resistance of the coil. It is minimised by using low resistance material like thick copper wire. AP

- Eddy current loss: - This is due to circulation of current induced in the core. AP

It is minimised by using a laminated core. AP

By using soft magnetic material - Hysteresis loss. AP

- Flux leakage. AP

Accept any two points explained accordingly

By using soft magnetic material - Hysteresis loss. AP

- Flux leakage. AP

By winding 2nd coil over 1st coil

Weighted score			6	5	4	4	4
EC _G	EX		7 and above	3-6	1-2	1	0
	AN		5 and above	2-4	1	0	1
EA _G	Weighted score		6	5	4		
	AP		3 and above	2	1		



UGANDA NATIONAL EXAMINATIONS BOARD

DRAFT

Subject:

Paper code: 535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year: 2025

EXPECTED RESPONSES

Item 7

7a(i) The missing components include:-

- A step down transformer / Adapter / Potential divider / charger
- Rectifier / Adapter
- Switches / Socket / extension
- Fuse / circuit breaker / Earth wire

Any four components

a(ii) - The step down transformer changes voltage from 240V to 12V to match the voltage rating for the torch and T.V

- The rectifier changes the transformed 12V for T.V and torch from a.c to d.c to confirm to the voltage requirement for the T.V and torch.

- The fuse prevents damage of the device in case of too much current by melting and disconnecting the circuit.

- The switch allows opening and closing the circuit which enables turning on and off of the gadgets independently.

Any four uses of each component

Notes: Score in use of component and component is considered correct if component is measured well



UGANDA NATIONAL EXAMINATIONS BOARD
DRAFT

19

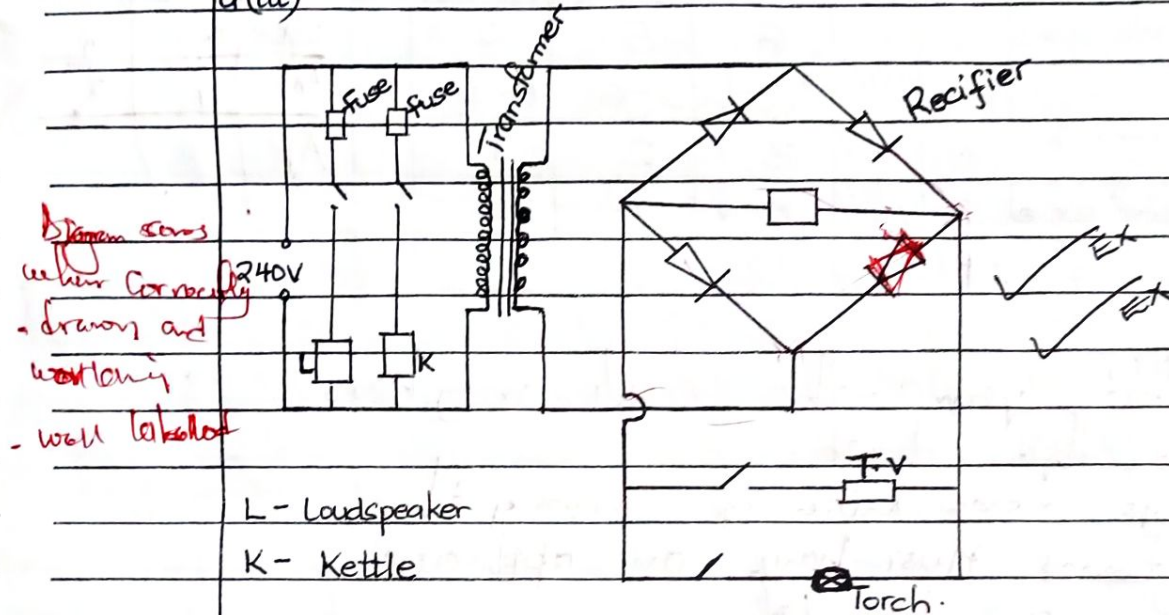
Subject:

Paper code:535/1

Level: ☐ PLE ☒ UCE ☐ UACE

Year:2025

a(iii)



Accept any correct diagram with components in parallel including a transformer and a rectifier

b) Amount of money needed to operate the loudspeaker

Handwritten notes:
- Commission
- Substitution
- values

$$\text{Amount} = \frac{600}{1000} \times 10 \times 7 \times 600 \quad \text{AN (substitution)}$$

$$\text{Amount} = 2520/- \text{ per week} \quad \text{AN}$$

Accept amount

$$= \frac{60}{1000} \times 10 \times 600 \quad \text{AN}$$

$$= \text{Shs } 2520 \quad \text{AN}$$

The business person can plan to spend 2520/- per week on the loudspeaker.

c) - Using thick/short good conducting cables. *AP*
This reduces resistance of the conducting cables hence reducing power loss during transmission (Power loss = $I^2 R$) *AP*

Handwritten notes:
- always to minimise + what it does
- to identifying purpose

- Use of energy saving devices. This ensures less power utilization when the devices are in operation. *AP*
- Switching off the appliances when not required. *AP*

